

Mathematical Preliminaries

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Mathematical Preliminaries

Topics

- Set Theory
- Combinatorics
- Growth Functions
- Sequences and Series

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Set Theory

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Set Theory

Definitions

A **Set** is a collection of objects called **members**. If a, b, c, d are members of a Set S we write : $S = \{a, b, c, d\}$

where $a, b, c, d \in S$

- Set **builder notation** can be used to define a set. For example, the set $S = \{1, 3, 5, 7, 11, 13\}$ can be defined as

$$S = \{x: \text{where } x \text{ is prime and } x < 15\}$$

- The **order** of members does not matter
 $S = \{a, b, c, d\} = \{d, b, c, a\}$

- All members are **distinguished** objects

- The **cardinality** of set is the number of elements. For example, the cardinality of S , denoted by $|S|$, is

$$|S| = 4$$

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Set Theory

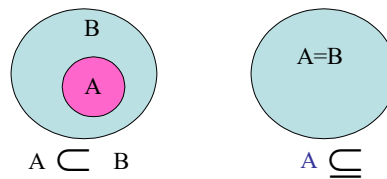
Subsets

A set **A** is called a **subset** of set **B** if every element of **A** is also a member **B**. Symbolically:

$$A \subseteq B, \text{ if } x \in A \text{ then } x \in B$$

The set **A** is **proper subset** of **B**, if **A** contains some of the elements of **B**. Symbolically:

$$A \subset B$$



Venn Diagram

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Set Theory

Universal Set

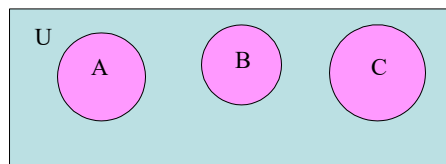
The largest set for a given problem is called **Universal** set. All other sets are **subsets** of the Universal set **U**. Symbolically:

$$\phi \subseteq A \subseteq B \subseteq C \subseteq U$$

Example: **U** is set of all small alphabets

$$U = \{a, b, c, \dots, x, y, z\}$$

$$A = \{a, f, g\}, \quad B = \{p, r, r, t\}, \quad C = \{x, y, w\}$$



Venn Diagram

Given a Universal set, we define **complement** A^c of set **A** as

$$A^c = U - A$$

Example: If $A = \{a, b, c\}$, then $A^c = \{d, e, f, \dots, x, y, z\}$

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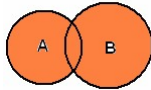
Set Operations

Union

The **union** of set **A** and set **B** is set of all elements x such that x is in **A** or x is in **B**. Symbolically:

$$A \cup B, \text{ if } x \in A \text{ or } x \in B$$

- The Union operation is pictorially represented by *Venn diagram*:



Example: $A = \{1, 3, 4, 5\}$, $B = \{3, 4, 7, 8, 9\}$. The Union of sets **A** and **B** is the set $\{1, 3, 4, 5, 8, 9\}$.

Example: Insert elements of sets $\{a, b\}$, $\{b, c\}$ into an empty set **S**.

Initializing S with empty set $\varnothing$

$$S = \varnothing$$

Inserting elements of first set $\{a, b\}$

$$S = S \cup \{a, b\} = \{a, b\}$$

Inserting elements of second set $\{b, c\}$

$$S = S \cup \{b, c\} = \{a, b, c\}$$

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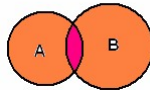
Set Theory

Intersection

The **intersection** of set **A** and set **B** is set of all elements x , such that x is in **A** and x is in **B**. Symbolically:

$$A \cap B, \text{ if } x \in A \text{ and } x \in B$$

- The intersection operation is pictorially represented by Venn Diagram.



Example: The Intersection of set $A = \{1, 2, 3, 4, 5\}$ and set $B = \{2, 4, 7, 8\}$ is the set $\{2, 4\}$

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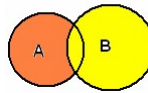
Set Operations

Difference Set

The **difference** of set **A** and set **B** is set of all elements **x**, such that **x** is **not** in **B** if **x** is in **A**.
Symbolically:

$$B - A, \text{ if } x \in A \text{ then } x \notin B$$

- The difference operation is illustrated by Venn Diagram.



Example: If $A = \{1, 2, 3, 4, 5\}$ and $B = \{4, 6, 8, 10\}$ then $B - A = \{1, 2, 3, 5\}$

Example: Delete elements a, d from the set $S = \{a, b, c, d, e, f\}$

Deleting a :

$$\begin{aligned} S &= S - \{a\} \\ &= \{b, c, d, e, f\} \end{aligned}$$

Deleting d :

$$\begin{aligned} S &= S - \{d\} \\ &= \{b, c, e, f\} \end{aligned}$$

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Set Operations

Properties

Disjoint Sets: Two sets **A** and **B** are said to be **disjoint** if

$$A \cap B = \varnothing$$

Distributive Properties:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

DeMorgan's Laws:

$$A - (B \cap C) = (A - B) \cup (A - C)$$

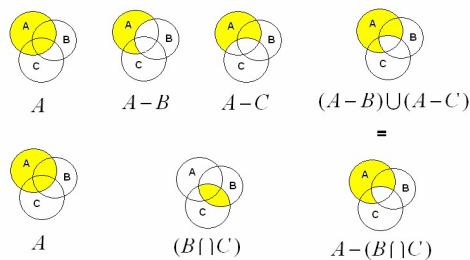
$$A - (B \cup C) = (A - B) \cap (A - C)$$

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Set Theory

DeMorgan's Law

$$A - (B \cap C) = (A - B) \cup (A - C)$$



Venn Diagram illustrating DeMorgan's Law

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Set Theory

Set Cardinality

The **cardinality** of a set is the number of elements in the set. It is also called **size**.
The cardinality of set S is denoted by $|S|$

Example: $S = \{a, b, c, d, e\}$, $|S| = 5$

Theorem: $|A \cup B| = |A| + |B| - |A \cap B|$

Example:

$$A = \{a, b, c, d, e, f\}, \quad |A| = 6$$

$$B = \{d, e, f, g, h, i, j, k\}, \quad |B| = 8$$

$$A \cup B = \{a, b, c, d, e, f, g, h, i, j, k\}, \quad |A \cup B| = 11$$

$$A \cap B = \{d, e, f\}, \quad |A \cap B| = 3$$

$$|A| + |B| - |A \cap B| = 6 + 8 - 3 = 11$$

Theorem: $|A \cup B| \leq |A| + |B|$

Equality is true for **disjoint** sets

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Set Theory

Cartesian Product

The **Cartesian product** of two sets **A** and **B**, denoted by **A X B**, is a set of all ordered pairs such that the first element of the pair is in set **A** and the second element is in set **B**. It is also called **cross product**. Symbolically:

$$A \times B = \{ (a, b) : a \in A \text{ and } b \in B \}$$

Example: Let $A = \{a, b, c\}$ and $B = \{g, h\}$, then

$$A \times B = \{(a, g), (a, h), (b, g), (b, h), (c, g), (c, h)\}$$

Theorem: Cardinality of Cartesian product is the product of cardinalities of sets

$$|A \times B| = |A| \cdot |B|$$

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Set Theory

Binary Relations

A **binary relation** R on sets **A** and **B** is a subset of the Cartesian product **A X B** of the sets. Symbolically:

$$R \subseteq A \times B$$

Example: Let $A = \{a, b, c\}$, $B = \{x, y\}$

$$A \times B = \{(a, x), (a, y), (b, x), (b, y), (c, x), (c, y)\}$$

$$R_1 = \{(a, y), (b, y), (c, y)\}$$

$$R_2 = \{(a, x), (a, y)\}$$

➤ A binary relation can be defined on a set **A** itself.

$$R \subseteq A \times A$$

Example: Let $A = \{a, b, c\}$

$$A \times A = \{(a, a), (a, b), (a, c), (b, a), (b, b), (b, c), (c, a), (c, b), (c, c)\}$$

$$R = \{(a, b), (b, c), (c, a)\}$$

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Combinatorics

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Permutations

Permutation

A **permutation** is an arrangement of n objects in some given order .

Elements of a set of n objects can be arranged in $n!$ (**n factorial**) ways
 where $n! = n.(n-1).(n-2).....3.2.1$

Example: Set = { a, b, c } has

$3! = 6$ permutations:

$abc, acb, bac, bca, cab, cba$

- Permutations are commonly used in **brute force** and **probabilistic algorithms**

Example: A graph consisting of n vertices has $(n-1)!$ Hamilton's circuits.

For, the first vertex can be connected to $n-1$ vertices. Thus, there will be $n-1$ paths between all pair of vertices.

Each of the $n-1$ vertices can in turn be connected to $n-2$ vertices. Thus, there will be altogether $(n-1)(n-2)$ paths among each of the three vertices. Continuing, in this way, the total number of Hamilton's circuits is

$$(n-1)(n-2)(n-3).....2.1 = (n-1)!$$

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Permutations

Factorial Growth

The **factorial function**, $f(n)=n!$, is among the *most rapidly growing function*, as shown in the table

n	n!
2	2
4	24
8	40320
16	20922789888000
32	32 2.63130836933693e+35

- It can be shown that $(1000)!$ is an integer consisting of **2500 digits**!
- In asymptotic analysis, which involves large values of n , the $n!$ is computed by using **Stirling's approximation**:

$$n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n, \text{ where } e = 2.718$$

Example: What is the total time to compute *all tours* for a graph with 41 vertices, assuming one *microsecond* is required to compute one tour.

A graph with 41 vertices has $40!$ tours. Using *Stirling approximation* it can be shown that $40! = 8.16 \times 10^{47}$

$$\begin{aligned} \text{Total computation time} &= 8.16 \times 10^{47} \text{ microseconds} \\ &= 8.16 \times 10^{47} / (1000000 \times 60 \times 60 \times 24 \times 365 \times 1000) \\ &= 3.1 \times 10^{16} \end{aligned}$$

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Counting and Probability

k-Permutations

An arrangement of n objects taken k ($k \leq n$) objects at a time is called ***k-permutations***.

The number of k -permutations is given by :

$$n(n-1)(n-2)\dots(n-k+1) = n! / (n-k)!$$

Example: 2-permutations of set $\{a, b, c, d\}$, $(4!) / (4-2)! = 24/2 = 12$, are

ab, ac, ad, ba, bc, ca, cb, cd, da, db, dc

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Combinations

Definition

The arrangement of n objects taken k ($k \leq n$) at a time, *without regard to order*, is called ***k-combination***.

Example: The set $\{a, b, c, d\}$ has following 2-combinations:

ab, ac, ad, bc, bd, cd

Note that combination ab is the same as ba

- In general, the ***k-combinations of n*** objects is given by the formula:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

- The k -combinations are called ***binomial coefficients*** because of their occurrence in expansion of the binomial expression $(x+y)^n$

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$$

- Combinations are frequently encountered in the analysis of algorithms.

Example: The total number of subsets of an n -element set $S = \{a_1, a_2, \dots, a_n\}$ is 2^n .

Number of sub sets with 0 element = $1 = \binom{n}{0}$

Number of subsets with 1 element = $n = \binom{n}{1}$

Number of subsets with 2 elements = $\binom{n}{2}$

.....
Number of subsets with n elements = $1 = \binom{n}{n}$

$$\left. \begin{array}{l} \text{Number of sub sets with 0 element} = 1 = \binom{n}{0} \\ \text{Number of subsets with 1 element} = n = \binom{n}{1} \\ \text{Number of subsets with 2 elements} = \binom{n}{2} \\ \text{.....} \\ \text{Number of subsets with } n \text{ elements} = 1 = \binom{n}{n} \end{array} \right\} \begin{array}{l} \text{Total subsets} = \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} \\ = (1+1)^n \quad (\text{binomial theorem}) \\ = 2^n \end{array}$$

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Growth Functions

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Growth Functions

Algorithm Functions

➤The **behavior of algorithms**, for different ranges of problem input size, are often described by **standard mathematical functions** which are referred to as **growth functions**: The functions are classified as **polynomial**, **polylogarithmic**, and **exponential**.

Polynomial: Polynomial functions are positive powers of an integer
i.e $f(n) = n^c$ where c is some positive constant. Examples are: $n^{0.5}$, n , $n^{5.2}$, n^3 , n^{100}

Polylogarithmic Polylogarithmic functions are powers of logarithmic function,
i.e $f(x) = (\log n)^c$. Examples are : $(\log n)^{0.5}$, $\log n$, $(\log n)^{10.5}$

Exponential : Exponential functions are powers of a constant i.e $f(x) = c^{kn}$, where c and k are constants. Examples are: 2^n , 3^{2n} .

➤An algorithm is often referred to as **polynomial**, **logarithmic**, or **exponential** according as the **running time is expressed in terms of polynomial, logarithmic, or exponential function**. For example, an algorithm with running time $T(n) = 2^n$ is said to be **exponential algorithm** and algorithm with running time $T(n) = n^2$ is **polynomial algorithm**.

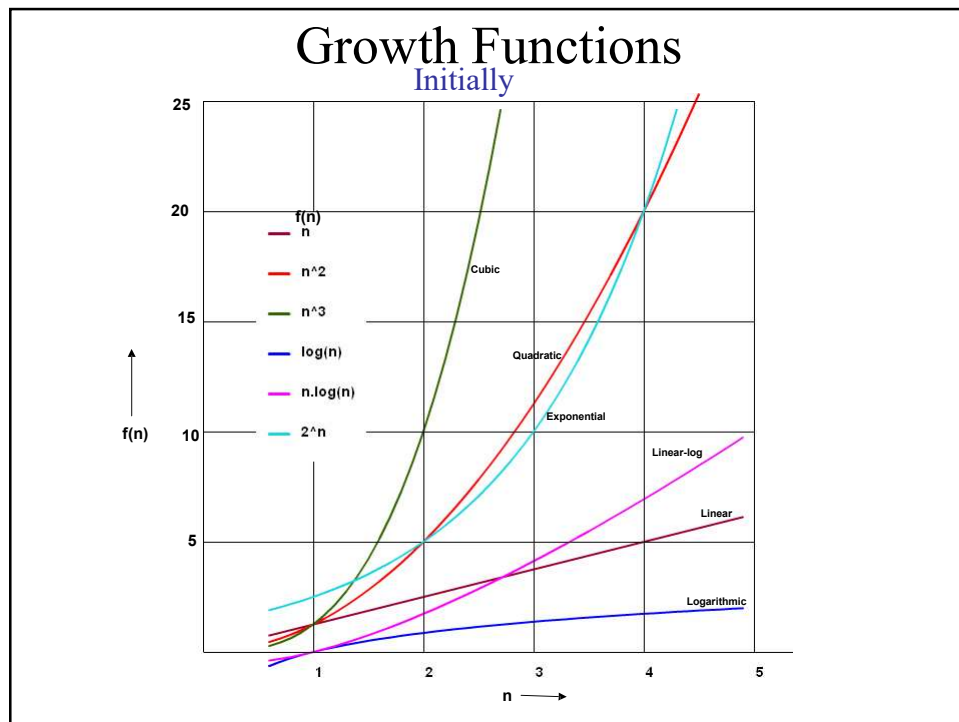
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Standard Growth Functions

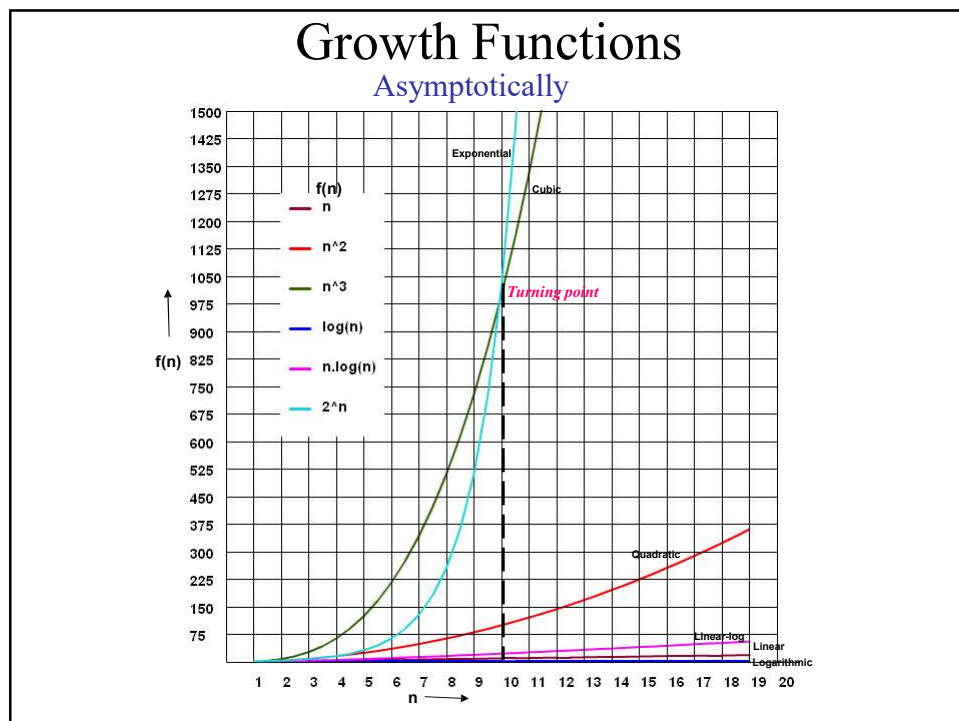
➤ Standard mathematical functions are often used in the analysis of algorithms
Their behavior for small input sizes is depicted in the table.

n	$\lg n$	n	$n \lg n$	n^2	n^3	2^n	$\sqrt{n}$
2	1	2	2	4	8	4	1.4
4	2	4	8	16	64	16	2
8	3	8	24	64	512	256	2.8
16	4	16	64	256	4,096	65,536	4
32	5	32	160	1,024	32,768	4,294,967,296	5.7
64	6	64	384	4,096	262,144	1.8×10^{19}	8
128	7	128	896	16,384	2,097,152	3.4×10^{38}	11
256	8	126	2,048	65,536	16,777,216	1.15×10^{77}	16
512	9	512	4,608	262,144	134,217,728	1.34×10^{154}	23

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Standard Growth Functions

Ranking

Algorithm	Class	Performance
$\log n$	<i>Logarithm</i>	<i>Very good</i>
n	<i>Polynomial Linear</i>	<i>Good</i>
$n \log n$	<i>n-logarithm</i>	<i>Fair</i>
n^2	<i>Polynomial Quadratic</i>	<i>Acceptable</i>
n^3	<i>Polynomial Cubic</i>	<i>Poor</i>
2^n	<i>Exponential</i>	<i>Bad</i>

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Comparison of Algorithms

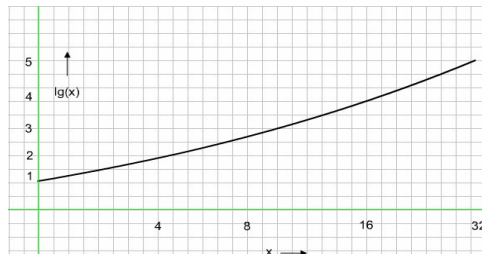
Processing speed : 1 millisecond

		Maximum Problem Size		
Algorithm	Time Complexity	1 Second	1 Minute	1 Hour
A1	n	1000	6×10^4	3.6×10^6
A2	$n \lg n$	140	4893	2.0×10^5
A3	n^2	31	244	1897
A4	n^3	10	19	153
A5	2^n	9	15	21

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Logarithm Function

If $y = \log_b x$, then $x = b^y$, b is called **base**.



➤ Logarithm is function often used in the analysis of algorithms
It is **slowly growing** function, and represents an **efficient algorithm**.

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Logarithm Function

Properties

➤ The following properties of logarithms are handy in the analysis of algorithms

- $\log_b x \cdot y = \log_b x + \log_b y$ (Sum rule)
- $\log_b x / y = \log_b x - \log_b y$ (Difference rule)
- $\log_b x^n = n \log_b x$ (Exponentiation rule)
- $\log_b x = \log_c x / \log_c b$ (Change of base rule)
- $\log_b x \log_b y = \log_b y \log_b x$ (Symmetry rule)

➤ Special notations are used for **base 2** and **base $e \approx 2.718$**

$$\log_2(x) = \lg(x) \quad (\text{Binary logarithm})$$

$$\log_e(x) = \ln(x) \quad (\text{Natural logarithm})$$

➤ Converting from binary algorithm to natural logarithm:

$$\lg(x) = \ln(x) / \log_e(2)$$

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Algorithm Functions

Floor Function

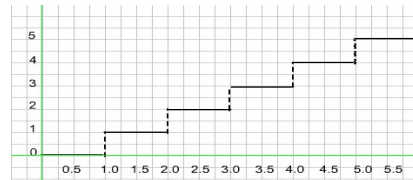
The **floor function** $y(x)$ is defined as:

$$y(x) = \lfloor x \rfloor$$

where y is the **largest** integer closer to x .

Example: $\lfloor 4.9 \rfloor = 4$, $\lfloor 4.0 \rfloor = 4$

It follows from definition: $x - 1 < \lfloor x \rfloor \leq x$



- Floor function grows **linearly** in steps
- Used for mapping **real valued** functions into **integer** functions

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Algorithm Functions

Ceil Function

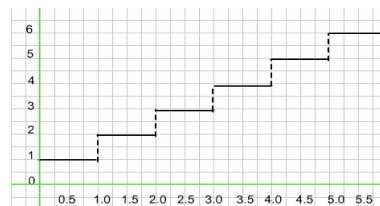
The **ceil function** $y(x)$ is defined as:

$$y(x) = \lceil x \rceil$$

where y is the **smallest** integer closer to x .

Example: $\lceil 4.9 \rceil = 5$, $\lceil 4.0 \rceil = 4$

It follows from definition: $x \leq \lceil x \rceil < x + 1$



- Ceil function grows **linearly** in steps
- Used for mapping **real valued** functions into **integer** functions

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Sequences and Series

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Sequences

A **sequence** is a function whose **domain** is the set of **natural numbers** $\{0, 1, 2, \dots\}$.

- The function $f(n) = n$, where $n = 1, 2, \dots$ defines the sequence $\{1, 2, 3, 4, \dots\}$, which called **arithmetic sequence**

- The function $f(n) = 2^n$ where $n = 0, 1, 2, \dots$ defines the sequence $\{1, 2, 4, 8, \dots\}$, which called **geometric sequence**

- Sometimes a sequence is defined by a **recursive relation**. An example is the celebrated **Fibonacci sequence**. It is defined by the relation

$$f(n) = f(n-1) + f(n-2), \quad \text{with } f(0) = 0 \text{ and } f(1) = 1$$

The recursion generates the sequence $\{0, 1, 1, 2, 3, 5, 8, 11, 13, 21, \dots\}$

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Series

Finite and Infinite

➤ A *series (summation)* is sum over a sequence. The series can be *finite* or *infinite*

If $S = \{a_1, a_2, a_3, \dots\}$ is *sequence*, the corresponding *finite* and *infinite series* are

Finite Series : $a_1 + a_2 + a_3 + \dots + a_n = \sum_{k=1}^n a_k$

Infinite Series: $a_1 + a_2 + a_3 + \dots + \infty = \sum_{k=1}^{\infty} a_k$

➤ Let $T(n)$ be the *sum of n terms* of a series.

Convergent Series: The series is called convergent if $T(n)$ tends to a *finite limit* when n tends to infinity. Symbolically, for a convergent series

$$\lim_{n \rightarrow \infty} T(n) = c, \text{ (} c \text{ is some constant)}$$

Divergent Series: The series is called divergent if $T(n)$ *tends to infinity* as n tends to infinity. Symbolically,

$$\lim_{n \rightarrow \infty} T(n) = \infty$$

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Series

Types

➤ The broad categories of series are::

- *Polynomial*
- *Exponential*
- *Harmonic*
- *Polylogarithmic*

➤ Series play an important role in the analysis of algorithms. They often arise in *iterative* and *recursive* procedures.

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Series

Polynomial

➤ Polynomial series are summations of **polynomial terms**, that is, terms consisting of n^i where n is an integer, and i is positive constant.

➤ Some common examples of **polynomial summations** are as follows. :

$$i=0, \sum_{k=1}^n 1 = 1 + 1 + 1 + \dots + 1 = n$$

$$i=1, \sum_{k=1}^n k = 1 + 2 + 3 + \dots + n = n(n+1)/2 \approx n^2/2$$

$$i=2, \sum_{k=1}^n k^2 = 1^2 + 2^2 + 3^2 + \dots + n^2 = n(n+1)((2n+1))/6 \approx n^3/3$$

$$i=m, \sum_{k=1}^n k^m = 1^m + 2^m + 3^m + \dots + n^m \approx n^{m+1}/(m+1)$$

➤ Symbol $\approx$ represents an approximation. Polynomial series arise in the analysis of **iterative algorithms**, such as selection sort, bubble sort

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Series

Exponential

➤ Exponential series are summations of **exponential terms**, that is, terms consisting of r^n where r is a constant, and n is the exponent. Series is also referred to as **geometric series**, and r is called **geometric ratio**

$$\sum_{k=0}^n r^k = (r^{n+1} - 1) / (r - 1) \quad (r \neq 1)$$

➤ The exponential series **converges** if $r < 1$, and **diverges** otherwise.

➤ The following exponential series, **with geometric ratio 2**, often arises in the analysis of **divide-and-conquer** algorithms.

$$2^0 + 2^1 + \dots + 2^n = 2^{n+1} - 1$$

➤ Another useful analysis series, referred to as **arithogeometric**, is

$$1.2^1 + 2.2^2 + 3.2^3 + \dots + n.2^n = (n-1) 2^{n+1} + 2$$

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Series

Harmonic

Harmonic series are summation of reciprocals of polynomial terms, that is, terms consisting of $1/n^i$, where $i=1, 2, \dots$ and n is a natural number.

$$i=1, \quad \sum_{k=1}^n 1/k = 1/1 + 1/2 + 1/3 + \dots + 1/n \approx \ln(n) + \gamma + 1/2n - 1/12n$$

$$i=2, \quad \sum_{k=1}^n 1/k^2 = 1/1^2 + 1/2^2 + 1/3^2 + \dots + 1/n^2 \approx \pi^2/6 - 1/n + 1/2n^2$$

$\gamma=0.57721$ is called *Euler's constant*, $\pi = 3.1416$

➤ The Harmonic series are frequently used in the analysis of sorting problems which use divide-and-conquer approach.

➤ There is no **closed** form for the summation of Harmonic series. Their **upper bound** and **lower bound** are obtained by using mathematical analysis, which replaces the **summation** by **integration**.